

CIFAR-10

$32 \times 32 \times 3$, 10 classes

50,000 training / 10,000 test images

2,000-image validation split held out before *any* training stage

ResNet-18 11.2M parameters

3×3 conv stem → 8 residual blocks (4 stages) → global avg. pool + FC (10 classes)

ViT-Tiny 5.7M parameters, trained from scratch

bilinear upsampling $32 \rightarrow 224$ → 16×16 patches 196 tokens + CLS dim 192 → 12 Transformer blocks, 3 heads, MLP ratio 4 → CLS head (10 classes)

Matched training recipe

stage 1 clean training (200 epochs)

stage 2 adversarial fine-tuning ($\leq$100 epochs, early stopping)

inner attack PGD-10 $\varepsilon = 8/255$, $\alpha = 2/255$

loss $(1-\lambda)\,\mathrm{CE}(x) + \lambda\,\mathrm{CE}(x_{\mathrm{adv}})$, $\lambda = 0.5$

3 independent runs per architecture

checkpoint: best PGD-10 accuracy on the held-out validation split

Evaluation full test set, per-sample logging ($n = 10{,}000$); the four protocols re-score the same records

- white-box robust accuracy PGD-10 and AutoAttack
- conditional decomposition robust = clean × conditioned survival
- transfer under four conditioning protocols raw ⊃ target-correct ⊃ both-correct ⊃ successful-source
- gradient structure sparsity, alignment, spatial locality
- depth profile block-wise feature drift, CLS attention entropy

Replications and controls. The whole design is repeated on CIFAR-100 (three runs per architecture) and SVHN (two runs); the CIFAR-10 checkpoints are re-evaluated under an L_2 budget ($\varepsilon = 0.5$); a RobustBench WideResNet-28-10 checkpoint enters only as the third model of the 3×3 transfer matrix; the checkpoint-selection rule is varied on a fixed training trajectory.